

Ethical A.I. Framework

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The adoption of A.I. is increasing by the day, and it is creating robust and scalable solutions that are exponentially increasing the productivity of businesses. With this increasing adoption, A.I. can potentially bring the next industrial revolution. With great power comes great responsibility, and A.I. is at its core just an extremely powerful tool. With that at one's disposal, ethical and moral standards are more important than ever, to maintain the social balance and power structure.

There are a few ethical guidelines that should be followed when building an A.I. for mass use. An example that is being deployed in masses is facial recognition cameras in public spaces, which aim to reduce crime rates in localities. We will be looking at this example and refer to some ethical guidelines that should be followed to make sure the deployment can adhere to the ethical standards and reduce the risk of destabilizing the society.

The following guidelines mentioned in the framework references the principles set in primarily 3 ethical frameworks for developments in A.I., as well as references from a few other frameworks. These are:

- i. European Union's trustworthy A.I. assessment framework
- ii. Canadian Federal Responsible A.I. Guidelines in Govt
- iii. Montreal Declaration: Responsible A.I. 10 principles

1. Privacy and Data Governance

Data privacy is one of the most important aspects of building a trustworthy business model that leverages artificial intelligence. The robustness, scalability, and accuracy of a machine learning model depend on the amount of correctly labelled data it has, and this data is collected over time to increase the accuracy. For our example use case of facial recognition, deploying A.I. powered cameras in public spaces does raise valid privacy concerns, and it can easily be used for mass surveillance. This data hence has to be well secured, anonymous, and private. The organization that is tasked with handling these vast amounts of data should have a robust data governance policy, to ensure the security and privacy of user data. Privacy and Data Governance is a major part of the European Union's trustworthy A.I. assessment framework.

2. Transparency

When it comes to complex machine learning models being deployed for the masses, transparency should always be a key aspect. When it comes to artificial intelligence, transparency relies on 3 major pillars, explainability, traceability, and communication. A machine learning model should be explainable regarding the parameters that its predictions are based on, the data sources should be traceable, and the organizations responsible for building and deploying these models should always have an easy and open communication channel open through which these factors can be communicated. For our example use case,

we should be able to explain how a detects a crime being committed, and also the training data should be traceable to ensure fairness in the process.

Transparency is one of the key pillars of ethical A.I. in most of the research frameworks that have been created, including the EU trustworthy A.I. assessment framework, the EY trusted A.I. framework and the Canadian Federal Responsible A.I. Guidelines.

3. Open-source

To increase fairness, security, and reliability, the implementation of A.I. should be open-sourced as much as possible, without compromising personal information. This means releasing the source code, training data, periodic internal test results, and any other relevant information. In our use case, this would mean releasing the source code for how these machine learning models are built, the training data, and periodic internal test results for accuracy, as well as data about when and how it is deployed. This would ensure that there are always more public eyes on the implementation, and ensure the security and privacy of individuals.

Being open-source is one of the key principles of the Canadian Federal Responsible A.I. Guidelines in Govt, and it is one of the major considerations when building ethical A.I.

4. Diversity, inclusion, and fairness

Machine learning models are as good as the data that it is trained on. These training data is collected, chosen, and labelled by humans, and humans by nature have some form of bias, which they inherit from society. In our example use case of facial recognition, ensuring the diversity of the training data, so that includes people of all races, gender, and ethnicity, as well as being fair when labelling the data should be a key priority. This can be done by having a diverse development team, as well as audits on the training data. Ensuring diversity, inclusion, and fairness will help protect human rights and maintain balance in society.

Diversity, inclusion, and fairness are described to be major pillars in all of the ethical frameworks derived from research in some form, the most notable ones being the EU's trustworthy A.I. framework, the Montreal Declaration: Responsible A.I. 10 principles, and more.

5. Accountability

Even though a machine learning model is technically a machine, how it works and makes decisions is ultimately at the hands of the developers. Accountability means that there are effective governance and oversight mechanisms for any project. Public accountability means that the public or its representatives are able to exercise effective oversight and control over the decisions and actions taken by the government and its officials, in order to guarantee that government initiatives meet their stated objectives and respond to the needs of the communities they are designed to benefit. Machine learning models by design are never perfect. For a use case like facial recognition and crime prevention, it puts a huge responsibility on the hands of A.I., which has the potential to destroy lives, it is extremely important to determine who is accountable for the predictions and decisions made by these models.

Accountability also is a core pillar of any ethical framework for responsible development in A.I, as described in the EU's trustworthy A.I. framework, UK data Ethics framework, and the Montreal Declaration: Responsible A.I. 10 Principles.